

1. Administrative & Study Identification

Full Study Title: A Phase 2, Open-Label, Multicenter Study of Arlocabtagene Autoleucel (BMS-986393), a GPRC5D-directed CAR T Cell Therapy in Adult Participants With Relapsed or Refractory Multiple Myeloma (QUINTESENTIAL) **Short Title/Acronym:** QUINTESENTIAL **Protocol Number:** CA088-1000 (Secondary ID: NCI-2024-02712) **NCT Number:** NCT06297226 **Posting ID:** 903223536626667746 **Trial Status:** RECRUITING **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Arlocabtagene autoleucel (arlo-cel / BMS-986393), a GPRC5D-directed autologous CAR T-cell therapy **Phase of Development:** Phase II **Medical Monitor:** Clinical Operations Lead, Bristol Myers Squibb / BMS Clinical Trials Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate Cohort 1: Best overall response (BOR) of partial response (PR) or better (the number and percent of participants achieving BOR of partial response (PR) or better in quadruple class exposed participants received at least 4 prior lines of therapy [LOT]). **Secondary Objectives:** To evaluate BOR of partial response (PR) or better; Best overall response (BOR) of complete response (CR) including stringent complete response (sCR); minimal residual disease (MRD) negative status; time to response; duration of response (DOR); progression-free survival (PFS); overall survival (OS); and safety.

2.2 Background & Rationale To address the significant unmet medical need in patients with relapsed/refractory multiple myeloma (RRMM) who are triple-class exposed (TCEx) and quad-class exposed (QCEx). While BCMA-targeted CAR T-cell therapies produce deep responses, many patients subsequently develop BCMA-negative disease and relapse. GPRC5D is an orphan receptor expressed on plasma cells, providing a novel and promising therapeutic target. Prior Phase 1 data demonstrated arlo-cel to be safe and highly efficacious in heavily pretreated RRMM patients, including those with prior exposure to BCMA-targeted therapy.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **N**: 230 participants (Targeted Enrollment) **Number of Sites**: Approximately 47 centers across the USA, Canada, and Japan **Estimated Study Start**: March 21, 2024 (First Patient First Visit) / July 08, 2024 **Estimated Primary Completion**: Patients will be followed for ≤ 5 years post-infusion, with a subsequent long-term follow-up study continuing for ≤ 15 years.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Documented diagnosis of multiple myeloma (MM) as per International Myeloma Working Group (IMWG) criteria. 3. Participants must have measurable disease during screening. 4. Received at least 4 classes of MM treatment (including an immunomodulatory drug [IMiD], proteasome inhibitor [PI], anti-CD38 mAb, and anti-BCMA therapy), and at least 3 prior lines of therapy (LOT). 5. Documented disease progression during or after their last anti-myeloma regimen as per IMWG 2016 criteria. 6. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

4.2 Exclusion Criteria 1. Active or history of central nervous system (CNS) involvement with MM. 2. Active systemic fungal, bacterial, viral, or other infection despite appropriate anti-infective treatment at the time of leukapheresis. Participants with severe infection, severe sepsis, or bacteremia in the last 28 days prior to leukapheresis are excluded. 3. Received any prior therapy directed at G protein-coupled receptor class C, group 5, member D (GPRC5D) or has received other prior treatment for MM without the required washout prior to leukapheresis. 4. Received an autologous transplant in the last 6 months, or an allogeneic transplant in the last year (or other protocol-defined criteria).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Leukapheresis followed by optional bridging therapy and lymphodepleting chemotherapy, followed by a single infusion of arlocabtagene autoleucel (target RP2D is typically 150×10^6 CAR T cells based on prior Phase 1 parameters). **Route of Administration**: Intravenous (IV) infusion. **Duration of Treatment**: Single one-time infusion, followed by short-term acute monitoring and long-term observation (≤ 15 years).

5.2 Concomitant & Prohibited Therapy Permitted: Systemic bridging therapy between leukapheresis and lymphodepletion to maintain disease control. **Prohibited**: Concurrent anti-myeloma therapies outside the study protocol; prior GPRC5D-directed treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, IMWG Disease Assessment, Leukapheresis
Baseline	Day -5 to 0	Lymphodepleting Chemotherapy
Treatment	Day 0	Single Infusion of arlo-cel, Acute Safety Monitoring (CRS/ICANS)
Interim	Day 29 / Q12W	Post-treatment disease assessment (IMWG), Safety Labs, MRD testing
End of Study	Year 5 / 15	Long-term follow-up for survival and secondary malignancies

7. Outcome Measures (Endpoints)

7.1 Primary Outcome Measure * Cohort 1: Best overall response (BOR) of partial response (PR) or better: The number and percent of participants achieving BOR of partial response (PR) or better in quadruple class exposed participants received at least 4 prior lines of therapy (LOT).

7.2 Secondary & Exploratory Outcome Measures * BOR of partial response (PR) or better. * Best overall response (BOR) of complete response (CR) including stringent complete response (sCR). * Minimal residual disease (MRD) negative status. * Progression-Free Survival (PFS) and Overall Survival (OS). * Duration of response (DOR) and Time to response. * General safety profiles.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection with specific focus on Cytokine Release Syndrome (CRS), Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS), hematologic toxicities (cytopenias), and on-target/off-tumor effects (e.g., skin, nail, and oral toxicities related to GPRC5D expression). **Serious Adverse Events (SAEs):** Reported within 24 hours. Participants must remain within a 30-to-120-minute transportation radius of the study site during the acute post-infusion monitoring phase (up to 1 month). **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms managed by Bristol Myers Squibb.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for primary efficacy; Safety Set for all patients receiving at least one dose of arlo-cel. **Sample Size Justification:** Target enrollment of 230 participants is statistically powered to evaluate meaningful efficacy (BOR of PR or better) and safety endpoints in a heavily pre-treated population. **Handling of**

Missing Data: Standard censoring methods via Kaplan-Meier estimates for time-to-event outcomes (e.g., PFS, OS, DOR).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: On-site and remote monitoring to ensure protocol adherence, safety tracking, and accurate reporting of response metrics via IRC. **Audit Trail:** eCRF locking and query resolution via standard clinical data management systems.

11. Study Identifiers & Governance

Primary ID: CA088-1000 **Posting ID:** 903223536626667746 **Registry Number:** NCT06297226 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** QUINTESENTIAL **Official Regulatory Title:** A Phase 2, Open-Label, Multicenter Study of Arlocabtagene Autoleucel (BMS-986393), a GPRC5D-directed CAR T Cell Therapy in Adult Participants With Relapsed or Refractory Multiple Myeloma (QUINTESENTIAL)

12. Clinical Framework (Multiple Myeloma Specific)

Disease Area/Condition: Multiple Myeloma (Relapsed or Refractory)

Diagnosis Threshold: Documented MM per IMWG criteria, refractory to at least ≥ 3 prior lines of therapy. **Medical Methodology:** GPRC5D-directed autologous CAR T-Cell Therapy **Optimization**

Target: Disease response (BOR of PR or better, CR including sCR) and MRD negativity.

13. Study Procedures & Methodology

Management Pathway: Leukapheresis $\rightarrow$ Bridging Therapy $\rightarrow$ Lymphodepletion $\rightarrow$ Infusion. **Education Protocol (HCP):** Specialized site training for handling and administering CAR T-cells, monitoring for CRS and ICANS.

Education Protocol (Patient): Strict monitoring mandates, requiring patients to reside close to the treating center with a dedicated caregiver for the first month post-infusion. **Quality**

Audit Mechanism: Continuous monitoring of product chain of identity and chain of custody.

14. Observation & Follow-Up Schedule

Total Duration: Up to 15 years (Long-term follow-up) **Onsite Visit**

Cadence: Intensive monitoring in the first 30 days, followed by

structured visits at Day 29, Month 3, 6, 12, etc. **Remote Monitoring Frequency:** Intermittent survival and subsequent therapy tracking during the long-term follow-up phase. **Checklist Review Interval:** Routine continuous assessment for delayed toxicities or secondary malignancies.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				